Joint residual encoding and a one-sample matroid prophet guarantee above 2^{-34}

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Abstract

We refine the analysis of OpenAI's one-sample matroid prophet construction against an adversary seeing all samples, values, and the entire random seed. A joint affine representation of two residual pivot tests reduces their combined counting dimension. A density-sample cutoff further reduces the residual family, with exceptional outcomes explicitly charged. Retaining the quadratic exit-hazard bound and recalibrating masks and finite phases gives a proposed guarantee above 2^{-34} with $37n+24$ initial fair bits. A compact exact coefficient is $1.055322895\ldots\,2^{-34}$, more than four times the preceding published certificate. This AI-assisted work has a written proof and same-agent verification. Independent review and formalization are pending; no priority or strongest-known-bound claim is made.

1 Statement and dependencies

Proposed theorem 1. *For each known finite labeled matroid M on n elements there is a measurable, distribution-independent online rule using one sample per label and $37n+24$ initial independent fair bits with the following property. Let all nonnegative sample/value coordinates S_e, V_e be mutually independent, with matching laws at each label and an independent seed. If $\mathbb{E}\,\mathrm{OPT}_M(V) < \infty$, then for every measurable arrival permutation depending on all samples, values, and the entire seed, the rule maintains independence at every prefix and earns expectation at least $C_1\mathbb{E}\,\mathrm{OPT}_M(V)$, where*

$$C_1 = \frac{14995972860899}{24412324730115194880000} > \frac{21}{20}\,2^{-34}.$$

We use the density maps, guarded paths, hybrid safety comparison, marked-pivot encoding, rank accounting, and lossless simulation in OpenAI's source [1], pinned at commit `fd4aeeb2ee4f` (full revision in the linked source). We specify the rule and source interfaces here, prove the new quadratic transaction and exit bounds, give the joint residual encoding and density cutoff, and derive the refined final constants. The structural matroid lemmas remain cited dependencies. No polynomial-time or bounded-precision claim is made.

2 The fixed-vector rule and its statistics

Fix a nonnegative hidden vector w before any seed bits. The rule reveals and sacrifices a seed-chosen mask and must handle every permutation; its target is $\mathbb{E}_R \min_\pi w(A(w,R,\pi))$. Write $\mathrm{r}(A \mid P) = \mathrm{r}(A\cup P) - \mathrm{r}(P)$ and $F_\varnothing = \mathrm{cl}(\varnothing)$. Fix a label order for every tie. Higher rounded weight means higher priority. Put $W = \mathrm{OPT}_M(u)$ and let I^* be priority greedy on the full rounded vector.

Use the parameters derived below:

$$\kappa = 640, \quad t = 2^{-15}, \quad \mathbb{P}(H_e) = 3/4, \quad \mathbb{P}(D_e) = 7/16, \quad \mathbb{P}(C_e) = \mathbb{P}(T_e) = t,$$

with main probability $1 - t$, fair parity, and uniform shared phase $J \in \{0, \dots, 255\}$. Put $\theta = 261/256$, $B = \theta^{256}$, and $u_J(x) = \theta^J B^{\lfloor \log_B(x/\theta^J) \rfloor}$ for $x > 0$, with $u_J(0) = 0$. All coins are independent and drawn initially.

The maximum branch reveals a separate fair mask and accepts the first unmasked positive nonloop with priority above every revealed positive nonloop. An empty threshold is below every positive nonloop. If m_* is the largest rounded nonloop weight, expected worst-order reward is at least $m_*/4$: the largest-priority nonloop is unmasked and the second is masked with probability $1/4$. With only one positive nonloop success has probability $1/2$; with none the bound is zero.

In the main branch reveal and sacrifice $H \cup D \cup C$. Define

$$U = \{e \notin H : u_e > 0,\ e \notin \mathrm{cl}(H_{>e})\}, \qquad Y = I^* \setminus H.$$

The candidate test uses only revealed H values and the tested value. For a fixed phase divide weights by θ^J, so true groups U_i have weight B^i. List those meeting D in increasing weight order as $G_1, \dots, G_m$; put $D_j = D \cap G_j$, and similarly C_j, T_j. The full groups are analysis objects; their revealed D_j, C_j parts suffice for preparation. If there are no listed groups, reject all.

For a flat P, let $Q_j(P)$ be the largest flat $Z \supseteq P$ maximizing $|D_j \cap Z| - \kappa\, \mathrm{r}(Z)$. Take closure first for nonflat arguments. Set $a_j = -3j$ and enable $Q_{j,k} = Q_j$ for $k \geq a_j$, with the identity earlier. Starting from $F_0(k) = F_\varnothing$ for $k < 0$ and E otherwise, define

$$X_j(k) = Q_{j,k}(F_{j-1}(k)), \qquad F_j(k) = \mathrm{cl}(X_j(k) \cup (C_j \cap X_j(k+1))).$$

Compute paths in increasing j on the finite relevant auxiliary timeline, and put $\widehat{F} = F_m$. For a candidate nonloop in listed group j, write $b_j(e) = \min\{k : e \in X_j(k)\}$. It is eligible precisely when it is unsacrificed, T-positive, has birth $b \geq a_j + 2$ of the chosen parity, and lies outside $\widehat{F}(b-2)$. At endpoint b maintain a set A_b; accept an eligible e exactly when $\mathrm{r}(\{e\} \mid \widehat{F}(b-2) \cup A_b) = 1$.

The source density and path lemmas give

$$F_{j-1}(k) \subseteq X_j(k) \subseteq F_j(k) \subseteq X_j(k+1), \qquad |D_j \cap (\mathrm{cl}(Q_j(P) \cup Z) \setminus Q_j(P))| \leq \kappa\, \mathrm{r}(Z \mid Q_j(P)).$$

Nonloops are born at a_j or in $[a_j + 2, 0]$, and the entire nominal path uses a generator set of size at most $|D_j|/\kappa$. The nested layer bases give prefix independence. Finite rank operations, fixed tie rules, and countably many rounding levels give measurability. These structural facts are inherited from the cited source.

For valid selected-parity endpoints $a_j + 2 \leq b \leq 0$, define

$$\begin{aligned}
K_{j,b} &= \bigcup_{\ell<j} (T_\ell \cap (X_\ell(b) \setminus X_\ell(b-1))), \\
P_{j,b} &= G_j \cap (X_j(b) \setminus X_j(b-1)), \quad O = D \cup C, \quad S_{j,b} = P_{j,b} \setminus O, \\
z_j &= \sum_b \mathrm{r}(S_{j,b} \mid X_j(b-2) \cup K_{j,b}), \\
\lambda_j &= \sum_b \mathrm{r}(T_j \cap S_{j,b} \mid \widehat{F}(b-2) \cup K_{j,b}).
\end{aligned}$$

The safe count σ_j counts $d \in D_j \cap Y$ with such a birth and $d \notin Q_j(F_{j-1}(b-2) \cup K_{j,b})$. For each nominal summand choose a fixed-order basis extension Z_b, so $\sum_b |Z_b| = z_j$. The residual is

$$\mathcal{R}_j = \bigcup_b (P_{j,b} \setminus \mathrm{cl}(X_j(b-2) \cup K_{j,b} \cup Z_b)) \subseteq O.$$

Extend σ_i, z_i, λ_i by zero on unlisted true groups. The source density argument gives $\sigma_j \le \kappa z_j + |\mathcal{R}_j|$. Its order-uniform lemma bounds the number of accepts of weight at least that of G_j below by λ_j, for every seed and order. One later-path generator budget over the whole timeline is

$$c_j = |C_j| + \sum_{\ell > j} (|C_\ell| + |D_\ell|/\kappa).$$

Conditional on H, D, C, parity, and test bits outside the focal group, independent test thinning gives $\mathbb{E}[\lambda_j \mid \mathcal{F}_{-j}] \ge t(z_j - c_j)$. These are the source interfaces used below.

3 A quadratic transaction inequality

Lemma 1. *Let $\mathcal{A}$ be an increasing event on finitely many independent bits, and let a predictable query procedure, reading each bit at most once, determine whether $\mathcal{A}$ occurs. Write $p = \mathbb{P}(\mathcal{A}^c)$. At the completed transcript let q be the failure probability with only unqueried bits random and every queried bit reset to zero. Then*

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] \ge p(1-p).$$

Proof. At an intermediate transcript, let W be its remaining coordinates and I its queried positives. Let m be the actual conditional failure probability, and q the failure probability with only W random and all other bits zero. Predictable product exposure identifies m with the probability when I is forced positive and W is random. Increasingness gives $0 \le m \le q \le 1$. Initially $m = q = p$.

For a predictably chosen next coordinate, the depleted probability q' is determined before the answer and satisfies $q' \ge q$. The actual conditional failure probability is a martingale: $\mathbb{E}[m' \mid \text{transcript}] = m$. Therefore the potential $\Phi(q, m) = q^2 - 2qm$ satisfies

$$\mathbb{E}[\Phi(q', m') \mid \text{transcript}] - \Phi(q, m) = (q' - q)(q' + q - 2m) \ge 0. \tag{1}$$

At the end $m = \mathbf{1}_{\mathcal{A}^c}$, and failure forces $q = 1$. The initial potential is $-p^2$; its final expectation is $\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] - p$. Sum (1) over the finite number of queries, padding stopped paths by zeros, to obtain the result. The proof includes $p = 0, 1$ and transactions needing no queries. □

4 The reverse exit hazard

Fix the source's filter and density masks, a listed-group prefix, and a nonloop d still alive before a reverse step. Its current states are S, and its transition is $\mathcal{G}_k(c, S)$. The remaining guard coordinates have their product law. Let $\mathcal{A}$ be the increasing event that d survives, and let p be the conditional exit probability.

For completeness, for $k < 0$ the source reverse map on a listed-group prefix is defined recursively by

$$A_0 = F_\varnothing, \qquad N_j = Q_{j,k}(A_{j-1}), \qquad A_j = \mathrm{cl}(N_j \cup \{e \in G_j \cap S_j : c_e = 1\}),$$

with output $(N_j)_j$. Starting from $X_j(0) = E$, reverse steps compute $X(k) = \mathcal{G}_k(C, X(k+1))$. For actual later states, $N_j \subseteq S_j$ for every fresh bit assignment. After computing N_j, query only $G_j \cap (S_j \setminus N_j)$; omitted additions already lie in N_j. The new remaining domain is $\bigcup_j (G_j \cap N_j)$.

The source queries exactly the departing coordinates in predictable batches. Their answers determine all output states: unqueried bits are inside flats already spanned. Thus the transcript decides $\mathcal{A}$, on both survival and failure, so Lemma 1 applies.

Let S' be the actual output and V' the remaining coordinates. For each assignment c supported on V', monotonicity in time and state gives

$$\mathcal{G}_{k-1}(c, S') \subseteq \mathcal{G}_k(c, S).$$

The right side has all just-queried coordinates zero, exactly the depleted experiment defining q. Conditional on the full post-step transcript, the bits in V' remain independent. On survival the next hazard r therefore satisfies $r \geq q$. The states are frozen in this comparison; earlier states are not recomputed after resetting bits. No order is conditioned on.

The next alive squared hazard consequently obeys

$$\mathbb{E}[\mathbf{1}_{\text{survive}} r^2] - p^2 \geq p - 2p^2.$$

Telescope this potential over the deterministic finite reverse horizon. Every nonloop exits, and if P is its hazard at actual exit, then

$$-p_{\text{start}}^2 \geq \sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i] - 2\sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i^2] = 1 - 2\mathbb{E}P.$$

Hence

$$\boxed{\mathbb{E}P \geq (1 + p_{\text{start}}^2)/2 \geq 1/2.} \tag{2}$$

In particular, the source's total squared-exit mass is at least $1/2$. Its hybrid comparison still applies because guard and test rates agree. The same baseline-birth loss and fair parity give, after averaging a density mask of arbitrary rate ρ,

$$\mathbb{E}[\sigma_i \mid H] \geq \gamma |Y_i| - \frac{\rho n_i}{2\kappa}, \qquad \gamma = \rho/4. \tag{3}$$

This averages before conditioning on listing, extending the statistics by zero on unlisted groups.

The bound $1/2$ is sharp for the abstract hypotheses. Take N Bernoulli-t bits and repeatedly test the OR of the remaining suffix, querying until the first positive or until that suffix is exhausted. The next hazard equals the depleted current failure probability. With $a = 1 - t$, classification by the last positive (or none) gives

$$\mathbb{E}P = a^{2N} + t\sum_{j=0}^{N-1} a^{2j} = a^{2N} + \frac{1 - a^{2N}}{2 - t}.$$

For $t = 1/m$, $N = m^2$, this tends to $1/2$. This is an abstract limitation, not a hard matroid instance or a bound on the best prophet algorithm.

5 Joint residual encoding and a density cutoff

Fix H, parity, and all mask bits outside a true group U of size $n \geq \kappa$. Do not condition on its focal bits or listing. Its incoming path $F(k)$, competition sets K_b, index, and activation a are fixed. Write $S_d(N) = \sum_{i=0}^{d} \binom{N}{i}$. Suppose a path has density generators J and rank witnesses Z, of sizes j, z. A generator has insertion time $\tau(g) \in [a, 0]$, and

$$X(k) = \text{cl}(F(k) \cup \{g \in J : \tau(g) \leq k\}).$$

The source gives two marked streams. First, list $F(k)$ and then generators inserted at k, marking the block one exactly when k is a valid endpoint of the chosen parity. Second, start with the fixed $F(b_0)$ marked zero, where $b_0 \in \{a-2, a-1\}$ has that parity. At each subsequent parity endpoint b, insert K_b, Z_b with mark zero, then $F(b)$ and generators with $b-2 < \tau(g) \leq b$ with mark one. End at $b_1 \in \{0,1\}$ of that parity; set $K_b = Z_b = \varnothing$ at invalid endpoints.

The second stream has intermediate span $\mathrm{cl}(X(b-2) \cup K_b \cup Z_b)$ and endpoint span $X(b)$, since $K_b \subseteq F(b)$ and $Z_b \subseteq X(b)$. The first test selects valid births. For a label passing it, the second pivot lies in the same layer and has mark one exactly on the residual. Thus the intersection of the two mark-one sets is the residual.

Lemma 2 (Joint pivot count)**.** *For fixed J, Z, the number of residuals obtained from these linked streams is at most $3^j 4^{2j+z} S_{j+z}(2n)$.*

Proof. The source marked-pivot argument first bounds retained greedy bases of occurrences by 4^s when s specified movable occurrences are inserted into an old sequence spanning the matroid. To recall why, old-only greedy gives an ordered basis. Only at most s of its positions can be omitted after insertions: their conditional-rank drops for the full movable set sum to at most s. There are at most 2^s choices of omitted old occurrences and 2^s of retained movable occurrences. Apply this to the two streams with $s = j$ and $s = j + z$, giving 4^{2j+z} pairs of retained bases. Fix a pair. Each nonloop has a unique minimal support in each basis, by matroid submodularity, and its pivot is the last support member.

We represent both streams by the same $j+z$ variables. Label the entire ground set $0, \ldots, N-1$, put $m = N+1$, $T = 8m$, and $u_k = (k-a+2)T$. Give old $F(k)$ occurrences in the first stream absolute scores 2^{u_k+4m+e}, with sign equal to their fixed marks. In the second stream, the initial $F(b_0)$ has negative scores $-2^{u_{b_0}+4m+e}$; K_b has scores -2^{u_b+e}; and $F(b)$ has scores 2^{u_b+4m+e}. These are fixed before the focal masks.

Give each generator a variable $x_g = 2^{u_{\tau(g)}+6m+g}$. Its first score is $\eta_g x_g$, positive exactly at a valid endpoint. Its second score is $d_g x_g$, where $d_g = 1$ for the chosen parity and $d_g = 2^T$ otherwise. Multiplication by 2^T moves an insertion from $b-1$ into the endpoint b block. Give each witness $e \in Z_b$ a variable $y_e = 2^{u_b+2m+e}$, used only as the score $-y_e$ in the second stream. There are only three types for a generator:

$$(\eta_g, d_g) \in \{(1,1), (-1,1), (-1,2^T)\}.$$

The middle type includes early insertions of the chosen parity that are not valid births. Fix all types, at a cost of at most 3^j choices. Order every block by label. Second-stream generators can be reordered within their endpoint block because all have mark one; this preserves pivot marks and endpoint spans, and all retained bases are already counted.

In each stream the absolute scores are distinct powers of two in occurrence order. Every score dominates the sum of its predecessors. The sign of the sum over a support therefore equals its last member's mark. For fixed bases and types these are $2n$ affine functions of the same $j+z$ variables. Every realized mark pair is a strict sign pattern. The hyperplane-region bound gives at most $S_{j+z}(2n)$ patterns, proving the claim. Ground-set size and timeline length affect coefficients, not this dimension bound. □

Allowing all J, Z of size at most q gives a family fixed before the focal masks with

$$|\mathfrak{R}| \leq 192^q S_q(n)^2 S_{2q}(2n). \tag{4}$$

The preceding separate count was $4^{3q} S_q(n)^3 S_{2q}(n)$; the joint representation reduces the leading logarithmic entropy coefficient from five to four, with the new type factor fully included.

Choose an analysis cutoff $\rho < r < 1$, and set $q = \lfloor rn/\kappa \rfloor$. Call $|D \cap U| \le rn$ good. On a good outcome the source generator bound gives $|J| \le q$. If the nominal rank $z_i \le q$, its witnesses have size at most q too. If $z_i > q$, then $\kappa z_i > rn \ge |D \cap U| \ge \sigma_i$, so no residual error is needed. With $\alpha = r/\kappa$, the binomial-sum bounds, including $q = 0$, imply

$$\log |\mathfrak{R}| \le A_r n/\kappa, \qquad A_r = r\big(4\log(\kappa/r) + 4 + 6\log 2 + \log 3\big).$$

Indeed $\log S_q(n) \le \alpha n \log(e/\alpha)$ and $\log S_{2q}(2n) \le 2\alpha n \log(e/\alpha)$, with floors intact.

Define e_i as the residual size on listed good outcomes with $z_i \le q$, as n on bad outcomes, and zero otherwise. Pointwise, $\sigma_i \le \kappa z_i + e_i$. The mask $O = D \cup C$ has independent focal rate $p_O = \rho + (1-\rho)t$ under the original conditioning. Put $\lambda = -\log p_O$. The good residual tail is bounded by $\min(1, |\mathfrak{R}|e^{-\lambda x})$: drop the good-event intersection before the union bound, without asserting a conditional product law. Its integral is at most $(A_r n/\kappa + 1)/\lambda$.

Independently, Chernoff's bound gives

$$\mathbb{P}(|D \cap U| > rn) \le e^{-nd(r\|\rho)}, \qquad d(r\|\rho) = r\log\frac{r}{\rho} + (1-r)\log\frac{1-r}{1-\rho}.$$

This follows by Markov's inequality with exponential tilt $e^s = r(1-\rho)/(\rho(1-r))$ and moment $(1-\rho+\rho e^s)^n$. Combining the branches, $n \ge \kappa$, and (3) yields

$$\mathbb{E}[z_i \mid H] \ge \frac{\gamma|Y_i| - \epsilon n_i}{\kappa}, \qquad \epsilon = \frac{\rho}{2\kappa} + \frac{A_r + 1}{\kappa\lambda} + e^{-\kappa d(r\|\rho)}. \tag{5}$$

The cutoff is only an analysis device and does not restrict the theorem or add an algorithmic sampling requirement.

6 Parameters and final accounting

Choose

$$\kappa = 640, \quad t = 2^{-15}, \quad h = \mathbb{P}(H_e) = 3/4, \quad \rho = \mathbb{P}(D_e) = 7/16, \quad \gamma = 7/64.$$

Guards and tests have rate t; parity is fair. Set main-branch probability $p_{\text{main}} = 1 - t$. The maximum branch retains its separate fair mask. For rounding take $\theta = 261/256$, $L = 256$, $B = \theta^L$, and uniform $J \in \{0, \dots, L-1\}$, with

$$u_J(x) = \theta^J B^{\lfloor \log_B(x/\theta^J) \rfloor}, \quad u_J(0) = 0.$$

The source rule otherwise stays the same. All choices are initial and independent.

Take $r = 33/64$. Rational atanh-series enclosures with explicit tails give

$$A_r + 1 < 41/2, \quad \lambda > 413/500, \quad \kappa d(r\|\rho) > 11\log 2, \quad B > 140.$$

An upper bound on the error in (5) is therefore

$$\bar{\epsilon} = \frac{7}{32\kappa} + \frac{41/2}{\kappa(413/500)} + \frac{1}{2048} = \frac{335021}{8458240}.$$

For general filter rate h, the elementary source filter argument gives

$$\mathbb{E}_H u(Y) = (1-h)W, \quad \mathbb{E}_H u(U) = \frac{1-h}{h}\mathbb{E}_H u(\text{greedy}(H)) \le \frac{1-h}{h}W.$$

Indeed, condition on preceding memberships in rounded priority: the span test is fixed, and the probabilities for a candidate and a greedy selected sample are respectively $1-h$ and h times that test. At $h=3/4$ the two coefficients are $1/4$ and $1/3$.

Put $\beta=(B-1)/B$ and

$$L_0=\bar{\epsilon}+t\kappa+\frac{t\kappa+\rho}{B-1},\qquad s=(1-h)\left(\gamma-\frac{L_0}{h}\right).$$

The source's order-uniform rank comparison and independent test thinning give, for large groups,

$$\mathbb{E}[\lambda_i\mid H]\geq\frac{t}{\kappa}\left[\gamma|Y_i|-(\bar{\epsilon}+t\kappa)n_i-(t\kappa+\rho)\sum_{j>i}n_j\right].$$

The weighted higher-group sum is at most $u(U)/(B-1)$. Omitted groups $n_i<\kappa$ cost at most $\kappa m_*/\beta$ in comparison mass, where m_* is the maximum rounded nonloop weight. Consequently

$$\mathbb{E}_R\min_{\pi}\text{reward}_{\text{main}}\geq\frac{\beta ts}{\kappa}W-t\gamma m_*.$$

The maximum branch guarantees $m_*/4$ in worst-order expectation, and $(1-p_{\text{main}})/4=t/4\geq p_{\text{main}}t\gamma$, cancelling the negative term.

For a positive x, let $k=\lfloor\log_\theta x\rfloor$. Its rounded values over all phases are θ^{k-r} for $r=0,\dots,L-1$, once each. Since $\theta^k\geq x/\theta$, the geometric sum gives

$$\mathbb{E}_J u_J(x)\geq\frac{\beta}{L(\theta-1)}x=\frac{\beta}{5}x.$$

Apply this to a fixed true optimal independent set to obtain the same factor for expected rounded optimum. Thus the final coefficient is

$$C=\frac{p_{\text{main}}\beta^2ts}{5\kappa}.$$

Exact integer arithmetic gives $261^{256}>140\cdot256^{256}$, hence $\beta>139/140$ and

$$s>\frac{\gamma}{4}-\frac{1}{3}\left(\bar{\epsilon}+t\kappa+\frac{t\kappa+\rho}{139}\right)=\frac{23047361}{3527086080}>0.$$

Substitution gives $C>C_1>(21/20)2^{-34}$ as stated. Every scalar comparison is certified with exact rational arithmetic. Relative to the published

$$C_0=\frac{9652683394380719}{6375070236351523499016192 00},$$

the conservative certificate satisfies $C_1/C_0=4.056975824\ldots>4$.

The minimum over all permutations remains inside the seed expectation at every stage. The source's glued-vector simulation therefore transfers this coefficient losslessly to one sample per label. Conditional on the entire seed the glued vector has the product value law; ignored samples, sacrificed values, and unused coins may all still inform the adversary's order.

7 Resources and evidence

Per label draw $2 + 4 + 15 + 15 + 1 = 37$ fair bits for the filter, density mask, guard, test, and maximum mask. Shared bits are $8 + 15 + 1 = 24$ for the phase, branch, and parity, including unused-branch randomness. All new scalar constants and phase multipliers are fixed rationals. Rank/closure tables and flat enumeration may be exponential and remain charged preprocessing and storage. Observation precision and integer level indices are unbounded in the original exact-real measurable model. Rational input implementations can compare exact rational powers, with costs depending on bit lengths and exponents. No polynomial complexity is claimed.

The local exact checks cover 4,916 reverse-process quadratic resource and drift inequalities and 3,153 global inequalities on 170 small matroid/weight instances. Additional checks cover all 199 increasing Boolean events on at most four bits at two rates, 5,992 biased-filter instances, 560 selected-residual expectations, and 40 rational-rounding boundary cases. The abstract sharpness formula is checked by enumeration. These are finite evidence; the universal claim rests on the written argument and cited source dependencies.

New exact checks cover 45,824 residual encodings and 331,491 affine pivot signs over 170 matroid/weight instances, including early invalid births and shifted generators. They also cover 48 binomial tails, 120 floor/binomial-sum cases, and 24 new-phase boundaries. An explicit negative control rejects omitting the score multiplier. These finite checks do not establish the uniform joint-counting lemma.

The same-agent audit rechecked the query martingale, predictable depleted probability, both terminal outcomes, frozen-state containment, telescope, integrated tail at $n = \kappa$, changed mask factors, rounding, and simulation. The new same-agent pass also checked shared variables, all three types, fixed old scores, the arrangement dimension, and the cutoff conditioning. Independent review and formalization remain outstanding. Repository preparation does not constitute independent validation, a priority determination, or a confirmed best-known-bound claim.

8 Related work and development history

OpenAI's manuscript [1] derives a 2^{-293} coefficient in this model, stronger than its 2^{-310} headline. That derived coefficient is our documented baseline; the structural dependencies are identified above.

Daniel (@firsttimelifer) publicly announced a 2^{-47} guarantee on October 8, 2026 [2]. A supporting write-up was unavailable for comparison at the October 10 audit; we have not verified its assumptions or proof. We acknowledge it as an unverified announcement and do not use it as an established baseline.

This research effort progressed through 2^{-63}, 2^{-43}, 2^{-36}, and 2^{-34} on October 10, 2026. The first checkpoint refined source parameters, affine branch mixing, and finite rounding offsets. The second introduced a logarithmic hazard potential and tighter residual accounting. The quadratic potential and integrated residual bound gave the published 2^{-36} version. The present joint count and density cutoff refine that version, preserved at commit 669e97c. The dated archive preserves the intermediate proofs, certificates, and review status. These are stages of one research effort, not separate discoveries of the original constant-guarantee theorem.

References

[1] OpenAI, *One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary*, September 23, 2026. Pinned manuscript index.

[2] Daniel (@firsttimelifer), Public announcement, October 8, 2026. Unverified; supporting proof unavailable at the audit date.